Securing Microservice Environments through Automated DevSecOps and Continuous Runtime Assurance

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Abstract

Modern software systems increasingly adopt microservice architectures and containerized deployments. While this brings scalability and flexibility, it also introduces new security challenges: a larger attack surface, complex dependencies, and dynamic runtime behavior. Traditional security methods that only check applications at deployment time are insufficient; attacks may still occur through vulnerable libraries, misconfigurations, or compromised images.

To address these challenges, organizations are shifting toward **DevSecOps**: embedding security practices into the entire software development and operations lifecycle. However, existing approaches often stop at build-time checks and do not provide strong guarantees once applications are running in production.

This thesis aims to bridge that gap by combining secure development pipelines with **continuous runtime inspection and assurance**, creating a holistic security environment for microservice systems.

Research Questions

1. How can automated pipelines reduce the time and effort needed to secure microservice-based applications from development through deployment?

2. What strategies are effective for continuously validating the integrity of running applications and detecting malicious activity?

3. What trade-offs arise between improving security, maintaining system performance, and supporting developer productivity?

Objectives

- Design a security framework that integrates development, deployment, and runtime protection for microservices.
- Automate security checks and policies throughout the software lifecycle to minimize human error and delay.
- Implement continuous runtime assurance to detect and respond to attacks that bypass build-time defenses.
- Evaluate the approach using realistic case studies and simulated attack scenarios.

Methodology

The research will involve:

1. Building a representative microservice environment as a case study.
2. Designing and integrating an automated DevSecOps process that includes both build-time and runtime security layers.
3. Introducing controlled attack scenarios to test the framework's ability to detect and prevent compromises.
4. Measuring outcomes using metrics such as detection accuracy, response time, impact on system performance, and effect on developer workflows.

Expected Contributions

- A holistic security framework that unifies development, deployment, and runtime assurance for microservices.
- Insights into balancing security automation with system performance and usability.
- Practical recommendations for organizations aiming to adopt DevSecOps securely and efficiently.